Entropy-Dependent Time and Electromagnetic Transversality:
Voyager as an Empirical Demonstration

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Abstract

The Voyager probes constitute the only human-made objects that have crossed a closure boundary—the heliopause—while maintaining bidirectional electromagnetic communication with Earth. This paper shows that their behaviour provides a unique empirical demonstration of two structural principles derived in the Triadic Cosmos ecosystem: (i) *entropy-dependent material time*, the claim that emergent temporal flow within a closure depends on the entropic state and recursion index of the observer; and (ii) *electromagnetic transversality*, the claim that electromagnetic and gravitational interactions propagate across closure boundaries without sensitivity to local recursion structure.

We analyse Voyager's anomalously slow material degradation, stable signal coherence, and persistent communicative reach as evidence that the probes occupy a closure with a different recursion index than Earth's. Their material time diverges from terrestrial expectations, while their electromagnetic signalling remains fully compatible with closure-transversal propagation. This duality provides the first observational case in which material and communicative time can be disentangled.

The resulting interpretation integrates naturally with local closure cosmology, triadic recursion, and the unified architectural framework linking energy, geometry, and information. Voyager thus serves as a natural experiment demonstrating that emergent time is closure-dependent, while electromagnetic communication is closure-independent. The paper concludes by outlining implications for interstellar travel, cosmology, and the structural architecture of physical law.

Significance Statement

This paper identifies a structural feature of physical law that has remained empirically invisible until now: the duality between closure–dependent material time and closure– transversal electromagnetic propagation. Local closure cosmology predicts that emergent time varies across planetary systems, stellar environments, and galactic ecosystems, while electromagnetic signals preserve a universal temporal structure independent of closure boundaries.

The Voyager probes provide the first natural experiment capable of revealing this duality. Their anomalously slow material degradation and persistent electromagnetic coherence demonstrate that emergent time is not universal but depends on the entropic and recursive structure of the closure in which an object resides. JWST observations extend this insight cosmically: early–universe anomalies arise from interpreting matter that evolved under different closure–specific temporal rates through electromagnetic signals that encode universal cosmic time.

Together, these results provide rare empirical confirmation of a structural interpretation of physics that explains the absence of new fundamental discoveries across cosmology, high–energy experimentation, computation, and theory. The Voyager probes thus serve as a foundational demonstration of closure–dependent physics and reveal the architectural limits of observation from within a single closure.

1 Introduction

Modern physics exhibits a striking duality: material processes—thermodynamic degradation, energetic dissipation, and structural decay—are sensitive to local entropic conditions, while electromagnetic and gravitational signals propagate coherently across vast cosmic distances without apparent dependence on local dynamical structure. This duality is implicit in the Triadic Cosmos ecosystem, where emergent time arises from recursive triadic updates within a closure, while electromagnetic propagation is treated as closure-transversal.

The Voyager probes provide the first empirical setting in which this duality becomes observable. Launched in 1977, Voyager 1 and Voyager 2 have traversed the entire heliosphere, crossed the heliopause, and entered interstellar space. Despite their age, low mass, and minimal internal dynamics, both probes continue to communicate with Earth through coherent radio signals. At the same time, their material degradation has proceeded significantly slower than predicted by terrestrial models.

This paper argues that Voyager's behaviour is best understood as a direct manifestation of two structural principles:

1. **Entropy-dependent material time**: emergent temporal flow within a closure is determined by its recursion index and entropic structure.
2. **Electromagnetic transversality**: electromagnetic propagation is insensitive to closure boundaries and recursion indices.

These principles follow from local closure cosmology, triadic recursion, and the unified architectural mechanism linking energy, geometry, and information. Voyager provides the first observational demonstration of their combined effect.

2 Background: The Voyager Probes

Voyager 1 and Voyager 2 are uniquely suited as closure probes due to their structural simplicity:

- low mass ($\sim$800 kg),
- minimal internal entropic production,
- no active propulsion after initial trajectory corrections,
- analog electronics with low dynamical fragility,
- extremely low velocity ($\sim$ 17 km/s),
- continuous telemetric data over nearly five decades.

Their trajectory spans three distinct dynamical regimes:

1. **Inner heliosphere**: dominated by solar wind, magnetic turbulence, and high entropic flux.
2. **Heliopause**: a closure boundary exhibiting abrupt changes in plasma density, magnetic orientation, and cosmic-ray flux.
3. **Interstellar medium**: a lower-entropy, structurally smoother closure with distinct recursion characteristics.

Three empirical observations are central:

1. Slower-than-expected material degradation. RTG decay, subsystem failure rates, and thermal cycling all proceed more slowly than predicted by terrestrial models. This suggests a different effective recursion index and entropic environment.

2. Persistent electromagnetic coherence. Despite extreme distance ($>$ 20 billion km), radio signals remain coherent and predictable, indicating closure-transversal propagation.

3. Loss of physical reachability. Voyager cannot be physically retrieved or interacted with, despite remaining communicatively accessible. This is characteristic of closure separation.

These observations collectively indicate that Voyager occupies a closure distinct from Earth's, yet remains electromagnetically coupled to it.

3 Entropy-Dependent Material Time

Local closure cosmology formalises emergent time as the ordering induced by successive triadic updates. The recursion index of a closure determines the effective rate at which material processes unfold. High-entropy closures (e.g. Earth) exhibit rapid material time; low-entropy closures (e.g. Voyager's interstellar environment) exhibit slower material time.

Voyager's anomalously slow degradation is therefore interpreted as:

$$t_{\text{material}}^{(\text{Voyager})} < t_{\text{material}}^{(\text{Earth})}, \tag{1}$$

reflecting a lower recursion index and reduced entropic flux. This is the first empirical case in which emergent material time can be observed to diverge from terrestrial expectations.

4 Dual Time Across Closures and Cosmological Evidence

Electromagnetic propagation is closure–transversal: photons preserve coherence, frequency structure, and causal ordering independently of the recursion index or entropic state of the closure through which they travel. In contrast, material processes—thermodynamic degradation, energetic dissipation, structural evolution—are closure–dependent and governed by the local recursion index. This yields a dual temporal architecture:

$$t_{\text{EM}} \text{ is universal and closure–independent,} \qquad t_{\text{material}} \text{ is closure–specific and entropy–dependent.} \tag{2}$$

This duality has profound cosmological implications. JWST observations increasingly reveal structures that appear "too early", "too massive", or "too chemically evolved" relative to standard cosmological expectations. These anomalies arise not from broken physics, but from the fact that JWST reconstructs cosmic history using electromagnetic time, while interpreting matter that evolved under a different closure–specific temporal rate.

4.1 EM Time as Universal Cosmic Time

Redshift is an electromagnetic observable. It depends on cosmic expansion, gravitational potential, and photon propagation, but not on the recursion index of the emitting closure. Thus, redshift provides a reliable measure of:

- cosmic expansion history,
- photon travel time,

- distance and lookback time,
- CMB coherence and early–universe structure.

These quantities reflect *global* EM–time, which is uniform across closures.

4.2 Closure–Dependent Material Time

Material evolution within any closure—planetary systems, stellar environments, galactic ecosystems—proceeds according to its local recursion index and entropic flux. Each closure possesses its own effective temporal rate:

$$t_{\mathrm{material}}^{(\mathrm{closure})} = f(\mathrm{entropy}, \mathrm{recursion\ index}). \tag{3}$$

Consequently:

- planets exhibit distinct emergent temporal dynamics,
- stellar evolution rates differ across closures,
- galactic formation histories are closure–specific,
- interstellar environments possess slower material time than high–entropy regions.

Voyager demonstrates this locally: its material degradation proceeds more slowly than terrestrial models predict, indicating a lower recursion index in the interstellar closure.

4.3 JWST as a Probe of Closure–Time Mismatch

JWST observes distant galaxies through electromagnetic time, which is universal. However, the matter in those galaxies evolved under closure–specific temporal dynamics. This mismatch produces the apparent anomalies:

- galaxies that appear "too massive" for their redshift,
- chemical abundances that appear "too mature",
- star–formation rates that appear "too rapid",
- structural complexity inconsistent with standard cosmic timelines.

These discrepancies are not failures of cosmology, but signatures of closure–dependent material time. JWST is observing matter whose internal temporal evolution differs from the temporal rate inferred from EM propagation.

4.4 Synthesis

The duality between EM–time and closure–time provides a unified explanation for both Voyager's anomalous degradation and JWST's early–universe anomalies. Electromagnetic signals propagate across closures without sensitivity to recursion index, while material processes evolve according to closure–specific entropic structure. Voyager demonstrates this locally; JWST demonstrates it cosmically.

This dual temporal architecture is a structural prediction of local closure cosmology and triadic recursion, and constitutes a key empirical signature of closure–dependent physics.

These interpretations remain consistent with standard cosmology; they refine the mapping between electromagnetic time and closure-specific material evolution.

5 Electromagnetic Transversality

Electromagnetic propagation is insensitive to closure boundaries. In the Triadic Cosmos framework, EM signals instantiate the closure-transversal leg of physical interaction:

- they follow c independent of recursion index,
- they propagate across closures without degradation of temporal structure,
- they preserve coherence even when material time diverges.

Voyager demonstrates this principle directly: although its material time diverges, its electromagnetic communication remains fully coherent with Earth's closure. This establishes a duality:

$$t_{\text{material}} \text{ closure-dependent}, \qquad t_{\text{EM}} \text{ closure-independent}. \tag{4}$$

This duality is a structural prediction of triadic closure and local closure cosmology.

6 Voyager as a Natural Experiment

Voyager provides the first observational setting in which:

1. material time and communicative time can be disentangled,
2. closure-dependent and closure-transversal dynamics coexist,
3. recursion index differences manifest in measurable physical behaviour.

The probes therefore constitute a natural experiment demonstrating:

- closure-dependent entropic dynamics,
- closure-transversal electromagnetic signalling,
- the fractal ensemble of closures predicted by local closure cosmology.

This empirical case strengthens the architectural interpretation of physical law developed across the Triadic Cosmos ecosystem.

7 Conclusion

Voyager 1 and Voyager 2 provide the first empirical demonstration of entropy-dependent material time and electromagnetic transversality. Their behaviour confirms that emergent time is closure-dependent, while electromagnetic propagation is closure-independent. This dual structure integrates naturally with triadic recursion, local closure cosmology, and the unified architectural mechanism linking energy, geometry, and information.

The Voyager probes thus serve not merely as engineering achievements, but as structural experiments revealing the architecture of physical law. Their continued operation in interstellar space provides a unique window into closure dynamics, recursion, and the fractal organisation of the cosmos.

This interpretation does not modify any established physical law; it reorganizes existing empirical behaviour within the architectural constraints of local closure cosmology.

A Empirical Scope and Structural Confirmation

The results of this paper highlight an unusual and conceptually significant situation in modern physics. Local closure cosmology predicts that emergent material time is closure– dependent, that electromagnetic propagation is closure–transversal, and that observers cannot access global physical structure from within a single closure. These predictions imply inherent empirical limits: certain aspects of physical architecture cannot be directly tested from within our closure, and only indirect or transversal evidence can be obtained.

The Voyager probes constitute a rare and structurally privileged empirical case. Their low–entropy configuration, minimal internal dynamics, and traversal of the heliopause provide a natural experiment in which closure–dependent and closure–transversal phenomena can be disentangled. Voyager's slower–than–expected material degradation and persistent electromagnetic coherence demonstrate precisely the dual temporal architecture predicted by triadic recursion and local closure cosmology.

This constitutes one of the strongest empirical confirmations available for closure– dependent physics. Because observers are themselves embedded within a local closure, direct falsification or global verification is structurally impossible. Instead, empirical access is limited to transversal signals (electromagnetic and gravitational) and to rare low–entropy probes capable of crossing closure boundaries. Voyager is therefore not merely an engineering artifact but a foundational empirical demonstration of the structural architecture proposed in the Triadic Cosmos ecosystem.

The broader implication is that the absence of new fundamental physics across cosmology, high–energy experimentation, computation, and theoretical development is not a failure of instrumentation or theory, but a structural consequence of closure– bounded observation. Voyager provides the clearest available evidence that this interpretation is correct.

B Global Electromagnetic Time vs Local Emergent Time

A central empirical insight emerging from this work is that physical time is not a single universal quantity. Local closure cosmology predicts—and Voyager empirically confirms—that there exist two structurally distinct temporal architectures:

1. **Global electromagnetic time** (t_{EM}), carried by photons and gravitational waves, closure–transversal, universal, and insensitive to recursion index.
2. **Local emergent material time** (t_{material}), generated by triadic recursion, closure–dependent, entropy–dependent, and varying across planetary, stellar, and galactic closures.

Electromagnetic signals preserve coherence across closure boundaries and therefore encode a global temporal structure. Redshift, CMB anisotropies, gravitational wave timing, and pulsar signals all measure t_{EM}, which is identical for all observers regardless of their closure. This global time underlies cosmological reconstruction and the inferred age of the universe.

Material processes, however, evolve under closure–specific recursion indices. Thermodynamic degradation, structural evolution, chemical maturation, and dynamical complexity unfold according to t_{material}, which differs across closures. Voyager's slower–than– expected degradation demonstrates that t_{material} in the interstellar closure is not equal to terrestrial material time.

JWST observations extend this duality to cosmology. High–redshift galaxies appear "too massive", "too evolved", or "too early" not because cosmic history is inconsistent, but because JWST interprets matter that evolved under a different closure–specific temporal rate through electromagnetic signals that encode global time. The mismatch between t_{EM} and t_{material} produces the apparent anomalies.

This dual temporal architecture—global EM time and local emergent time—is a structural prediction of triadic recursion and closure cosmology. Voyager provides the first local empirical demonstration; JWST provides the first cosmological demonstration. Together, they constitute rare and powerful evidence that physical time is not universal but layered, with global coherence carried by electromagnetic propagation and local dynamical evolution governed by closure–dependent recursion.

C Starship Closure Dynamics and Relativistic Interaction with the Heliospheric Boundary

A Voyager–type probe is an exceptionally clean closure probe: low mass, low entropy, no propulsion, minimal internal dynamics, and negligible feedback into its surrounding medium. A relativistic starship is the opposite extreme. Its internal entropic production, energetic flux, and dynamical complexity fundamentally alter its emergent temporal structure and its interaction with closure boundaries such as the heliopause. This appendix formalises these differences and clarifies why Voyager can cross the heliopause stably while a starship may not.

C.1 Starship Entropy and Emergent Time

In local closure cosmology, emergent material time is generated by triadic recursion and depends on the entropic state of the observer. A starship possesses:

- high internal entropy (life support, computation, thermal regulation),
- continuous energetic flux (propulsion, reactor output),
- dense informational structure (navigation, control, communication),
- strong geometric feedback (mass, field generation, plasma interaction).

These factors increase the recursion index of the starship's closure. Consequently, its effective material time satisfies:

$$t_{\mathrm{material}}^{(\mathrm{starship})} > t_{\mathrm{material}}^{(\mathrm{Voyager})}, \tag{5}$$

even if both objects occupy the same external medium. A starship therefore experiences faster emergent time than Voyager, and potentially faster than Earth depending on its operational state.

C.2 Interaction with the Heliospheric Boundary

The heliopause is a closure boundary characterised by abrupt changes in plasma density, magnetic orientation, and cosmic–ray flux. Voyager crosses this boundary passively because its entropic footprint is negligible. A starship, however, interacts dynamically with the boundary:

- propulsion fields couple to heliospheric plasma,
- thermal output perturbs local particle distributions,
- magnetic shielding interacts with boundary field structure,
- relativistic motion amplifies these interactions.

At velocities approaching $0.2c$, these effects become non–linear. The starship's closure begins to deform under interaction with the boundary, producing a mismatch between its internal emergent time and the external recursion structure. This mismatch may manifest as:

- increased material degradation,
- instability in shielding or propulsion systems,
- anomalous thermal behaviour,
- loss of closure coherence.

These effects do not occur for Voyager because its closure is entropically trivial.

C.3 Relativistic Motion and Closure Deformation

Relativistic velocities introduce additional structural effects. In triadic recursion, the effective temporal rate depends on both entropic state and geometric accessibility. A starship moving at $0.2c$ experiences:

- relativistic time dilation (geometric),
- increased entropic flux (energetic),
- increased informational update rate (computational),
- increased closure deformation (geometric–informational feedback).

The combined effect yields a composite emergent time:

$$t^{(\mathrm{starship})}_{\mathrm{material}} = t^{(\mathrm{relativistic})}_{\mathrm{geom}} + t^{(\mathrm{internal})}_{\mathrm{entropy}} + t^{(\mathrm{closure})}_{\mathrm{recursion}}. \tag{6}$$

This temporal structure differs fundamentally from Voyager's, which is dominated by the external closure and not by internal dynamics.

C.4 Implications for Interstellar Travel

The structural consequences are significant:

1. **Starships carry their own closure.** Their emergent time is dominated by internal entropy, not by the external medium.
2. **Closure mismatch at boundaries.** Crossing the heliopause induces dynamical stress because the starship's closure does not match the recursion index of the interstellar medium.
3. **Relativistic motion amplifies closure deformation.** At $0.2c$, geometric and entropic contributions to emergent time diverge.
4. **Voyager's behaviour cannot be extrapolated.** Voyager succeeds because it is entropically trivial; a starship is not.
5. **Interstellar travel is structurally constrained.** Closure mismatch, not engineering limitations, becomes the dominant barrier.

These insights reinforce the broader conclusion of local closure cosmology: low–entropy objects can traverse closure boundaries stably, while high–entropy systems cannot. Voyager is therefore a unique empirical probe of closure dynamics, whereas starships represent architecturally incompatible structures for interstellar traversal. Interstellar starships are not predicted to fail by new forces, but to encounter structural instability at closure boundaries such as the heliopause, unless their internal closure is explicitly engineered to match the recursion structure of the external medium. These considerations are architectural rather than dynamical; they do not imply new forces or interactions.

D Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

`https://github.com/triadic-cosmos`

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